

Beyond Keywords: Comparing LLM-Assisted Topic Modeling Approaches Across Political Communication Contexts

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Abstract

This study provides a comprehensive evaluation of large language model (LLM)-assisted topic modeling across diverse political communication datasets, proposing a conceptual distinction between topic categorization (manifest subject mapping) and concept induction (latent construct discovery). We compare four methods, LDA, BERTopic, TopicGPT, and LLoOM paired with four proprietary LLMs and one open-weights model across social media posts, party platforms, and legislative bills. Using a multi-dimensional framework of alignment, coherence, diversity, and human interpretability, we find that TopicGPT excels at high-precision topic categorization, achieving peak semantic alignment especially in structured policy contexts. Conversely, LLoOM functions as a powerful tool for concept induction; although it may diverge from manifest ground-truth labels, it frequently demonstrates stronger text-topic relevance by capturing latent discursive logics that deductive approaches may overlook. Our findings further reveal that GPT-family models generally mitigate semantic vagueness, whereas the open-weights model shows a distinctive capacity for thematic diversity and performs competitively in some alignment settings, though its performance varies across contexts. Human annotation uncovers a semantic compression loss, where models identify rich latent themes but struggle to condense them into concise labels, highlighting the persistent need for human interpretive oversight. We conclude by offering an epistemological and practical framework to guide communication scholars in selecting GenAI-assisted methods that align with their theoretical level of meaning.

Keywords: Topic modeling, Concept induction, Generative AI, Large Language Models

Introduction

Topic modeling is a widely used method in communication research for analyzing unstructured and extensive textual corpora, drawing on textual data such as social media posts, newspaper articles, academic abstracts, and more (Guo et al., 2016; Rains et al., 2020). Topic modeling serves as a powerful tool for exploratory data analysis, condensing large volumes of text into a manageable set of topics or themes through dimensionality reduction without requiring training data. It facilitates the discovery of hidden patterns that may be challenging to detect through human coding or prior knowledge (Nelson, 2020).

Topic modeling has been widely applied by communication scholars across different subfields. For instance, in political communication, it is used to examine salient themes in public discourse around elections (Walter & Ophir, 2021), policy debates (Rossiter, 2022), and news events (Ghosh et al., 2022). In health communication, topic modeling helps identify emerging concerns or misinformation trends in online discussions about vaccines (Peng & Wang, 2024) or mental health (Shan et al., 2022). In interpersonal communication, it has been used to examine communication between parents and children (Baumer et al., 2024) as well as dynamics in romantic relationships (Moore & Ophir, 2022). The widespread adoption across communication subfields demonstrates topic modeling's versatility as a methodological tool.

Despite its widespread adoption, topic modeling continues to face concerns regarding validity and evaluation. For instance, a systematic review of 789 studies using topic modeling finds no convergence toward standardized validation practices and highlights a persistent tension between the inductive, exploratory nature of topic modeling and the deductive, quantitative tradition that favors clear validation criteria (Bernhard-Harrer et al., 2025). There exists an absence of shared guidelines, raising questions about how researchers should assess topic quality, interpretability, and theoretical alignment. At the same time, recent advances in large language models (LLMs) suggest new possibilities for improving computational text analysis. LLMs have potential in enhancing accuracy, flexibility, and semantic understanding in text classification tasks (Chae & Davidson, 2025; L. Li et al., 2024).

While recent research has focused on embedding LLMs into topic modeling pipelines, few studies offer comparative evaluations to assess their performance on general tasks, leaving a gap that the present study seeks to address. This study builds on recent advancements by comparing two

state-of-the-art LLM-assisted topic modeling frameworks, *TopicGPT* and *LLooM*, alongside the baseline model *LDA* and *BERTopic*, to address enduring challenges of interpretability in topic modeling. We examine how these models function in applied settings and how their distinct design philosophies may serve different analytical purposes within communication research. By evaluating both frameworks across multiple datasets and LLMs, this study offers a practical guide for communication scholars aiming to leverage LLMs effectively. Our goal is to help researchers navigate the trade-offs between automation and interpretability, while promoting a more transparent and theoretically grounded approach to computational topic analysis.

Literature review

Pre-GenAI Topic modeling

Topic modeling methods are conventionally grounded in probabilistic frameworks. A “topic” in topic modeling refers to a group of words that often appear together in the same kind of text (Blei, 2012; Blei et al., 2003). For example, a “political communication” topic mostly includes words related to politics and media, while a “health communication” topic mostly includes words related to health messages and public health. One of the early milestones in topic modeling is the LDA model (Blei, 2012; Blei et al., 2003). LDA assumes that each document is made up of several topics, and each topic is made up of certain words. It also assumes that the way topics are spread across a document, and the way words are spread within a topic, both follow a special kind of probability distribution called a Dirichlet distribution. The Dirichlet distribution is used because it directly produces valid probability mixtures (values between 0 and 1 and add up to 1) without any extra transformation, making it ideal for representing how topics mix in a document or how words mix in a topic. LDA has a wide application in the social sciences, including but not limited to analyses of survey responses, interview transcript, and social media posts (Churchill & Singh, 2022; Munro & Ng, 2022; Y. Wang et al., 2012).

The 2000s and early 2010s saw rapid development in topic modeling through various extensions of LDA, as researchers adapted the model to incorporate document structure, metadata, temporal dynamics, and supervised labels (Blei, 2012). LDA excels in identifying word co-occurrence patterns across documents. It is well-suited for identifying themes that are jointly discussed within the same documents. However, this design choice

also implies that LDA captures distributional proximity rather than semantic relatedness. As a result, semantically related words, such as “vote” and “ballot,” may be assigned to different topics when their usage patterns are systematically partitioned across documents, genres, or discursive communities, because LDA clusters words based on shared document distributions rather than conceptual similarity. While larger corpora increase the chance that related terms co-occur, LDA’s reliance on document-level frequency patterns means that semantic similarity is captured only insofar as it is reflected in shared distributional structure. Moreover, recent critiques and empirical evaluations of classical co-occurrence models highlight additional, systematic drawbacks. For instance, for corpora composed of short or sparse texts, co-occurrence information may be too weak to support stable topic inference, leading to incoherent or trivial topics (Egger & Yu, 2022). In addition, scholars criticize the widespread use of probabilistic models without adequate validation or human-centered interpretability checks, especially in social science contexts where theoretical relevance matters (Meaney et al., 2023).

Word embeddings, especially word2vec (Mikolov, Chen, et al., 2013; Mikolov, Sutskever, et al., 2013), focus on representing how similar words are in meaning, even when those words do not frequently appear in the same documents. Instead of relying on document-level co-occurrence, word embeddings represent each word as a vector in a continuous space, such that semantically similar words are located near each other. For instance, the words “vote” and “ballot” may be represented by nearby vectors, indicating that they are used in similar contexts and thus have related meanings. In practical applications, these embeddings are typically high-dimensional, but the underlying logic remains the same: words with similar meanings are positioned closely in the vector space. Building on this idea, embedding-based topic models became increasingly prominent in the late 2010s and early 2020s. One representative example is the Embedded Topic Model (ETM), which integrates word embeddings into topic modeling through a neural variational framework (Dieng et al., 2020). Unlike LDA, ETM does not assume a Dirichlet prior over document-topic mixtures. Instead, it uses a logistic-normal distribution and applies variational inference to learn topic proportions in the same semantic embedding space, allowing the model to capture more meaningful relationships among words and topics.

Despite the advances of topic modeling, from the probabilistic foundations of LDA to embedding-based models like Embedded Topic Model, some core challenges remain. First, topic modeling typically generates a collec-

tion of keywords to represent topics. For example, [scores, won, team, goals, game, offense, cup, boston, division, york], which might be interpreted as a topic related to *sports*. However, interpreting the meaning of each keyword group is a task for humans. In other words, topic modeling provides clusters of related terms, but it is up to the researcher to assign meaning to these clusters. While some studies rely on qualitative approaches such as “eyeballing”, “expert review”, “crowd-sourcing” to interpret topics, the interpretation process remains inherently subjective (Chen et al., 2023; Ying et al., 2022). Second, there is often no objectively “right” number of topics (Chen et al., 2023; Grimmer et al., 2022). Different choices can reveal different structures in the data. A model with 10 topics might give broad themes, while one with 50 topics might offer finer detail, but both could be useful depending on the research question. Third, topic modeling is often sensitive to initialization. Different starting points can lead to different final results because the algorithm might settle into different “local optima”, which might seem good within their neighborhood but are not necessarily the best possible arrangement across all possibilities (Blei et al., 2003; Chang et al., 2009). Fourth, exact replication of topic modeling can be challenging, even when using the same data and setting a fixed random seed. This is because some topic modeling tools make small decisions that can vary across computers or software versions (Bergstra et al., 2013). These differences can add up during the many steps of the algorithm, leading to slightly different results. So, while the general themes often stay similar, the exact output may not be identical across runs. Taken together, it is recommended that communication researchers approach topic modeling not as a purely objective tool, but as a flexible method that requires careful judgment.

Recent developments in GenAI-driven topic modeling

In the rapidly evolving era of Generative Artificial Intelligence (GenAI), LLMs have shown great potential in supporting automated content analysis of text-based data (Bail, 2024; L. Li et al., 2024; Ziems et al., 2024). Over the past two years, LLMs have greatly advanced traditional topic modeling tools by enhancing semantic coherence, improving embedding quality, and increasing interpretability. In the following section, we review recent LLM- and GenAI-based topic modeling methods and examine how well they respond to the limitations of earlier approaches. In the review below, we use “LLM” to refer to algorithms trained on massive amounts of text data to perform a wide range of language understanding tasks, often containing hundreds of millions or more parameters.

LLM-based embedding

Unlike older methods that give each word a single fixed vector, LLMs produce contextual embeddings, meaning the same word can have different vectors depending on how it's used. For instance, BERTopic (Grootendorst, 2022) uses Bidirectional Encoder Representations from Transformers (BERT) to generate embeddings. BERT was developed by Google in 2018 (Devlin et al., 2019). Like word2vec, BERT produces numerical embeddings. But unlike those older models, BERT considers surrounding words, so the same word can have different embeddings depending on context (Churchill & Singh, 2022). For example, “bank” in “river bank” and “money bank” will have different embeddings with BERT. BERTopic then groups documents with similar meanings based on their embeddings. It identifies clusters of related texts, which are then used to define distinct topics.

LLM-generated embeddings offer a key advantage: they capture the contextual meaning of words or sentences based on how they are used in specific passages of text. BERTopic uses such embeddings to reflect subtle shifts in meaning depending on surrounding words, leading to more accurate and semantically rich representations for downstream tasks such as topic modeling, clustering, and classification. However, although these embeddings improve semantic representation, they do not by themselves resolve the interpretive challenge of assigning clear meanings to topics

Meaning extraction

One of the long-standing challenges in topic modeling is that, while models can generate clusters of related words, it is still up to the human researcher to decide what those clusters actually mean (Chen et al., 2023). LLMs offer a promising way to support this interpretation process: when prompted appropriately, they can generate natural-language summaries or labels for topics, enhancing both the usability and accessibility of topic modeling. This does not imply that LLMs offer an objective or inherently superior interpretation of topics compared with human researchers. Like human readers, LLMs rely on prior textual exposure and contextual framing, and their outputs are shaped by training data, prompts, and modeling choices. LLMs may standardize or externalize parts of the interpretive process, making topic labeling more transparent, reproducible, or scalable across large numbers of topics. Nonetheless, final interpretation and theoretical grounding remain the responsibility of the researcher.

Recent studies have explored the use of LLMs to support topic represen-

tation and interpretation across various frameworks. Building on BERTopic (Grootendorst, 2022), Invernici et al. (2025) leverages LLM-based embeddings to improve document representations. Other approaches incorporate LLMs to move beyond word-level co-occurrence. For example, *PromptTopic* (H. Wang et al., 2023) uses prompting to capture sentence-level semantics, while Kapoor et al. (2024) employs LLMs to extract key phrases for more flexible topic constructions. *TopicGPT* (Pham et al., 2024) introduces an iterative, prompt-based approach to generate and refine topics. The *LLooM* framework (Lam et al., 2024) shifts from keyword-based topics to higher-level, theory-informed concepts with explicit inclusion criteria, reducing the reliance on extensive human interpretation and addressing the challenges highlighted by Chen et al. (2023). *LLooM* measures how well a text aligns with specific concepts by providing concept scores for each document. Compared with BERTopic, *LLooM* demonstrates superior concept coverage and generates more nuanced, high-quality topics.

These innovations mark a significant departure from the traditional bag-of-words approaches, but they also raise new questions about scalability, coherence, and human interpretability. While BERTopic remains favored among qualitative researchers for its accessible and coherent topic clusters (Kaur & Wallace, 2024), LLM-based frameworks like TopicGPT and LLoOM offer deeper, more semantically rich alternatives. Importantly, both models integrate LLMs not just for labeling but for structuring topics in a way that aligns with human reasoning through quotations, conceptual definitions, or explicit inclusion criteria, to address longstanding concerns about topic quality and interpretability (Z. Li et al., 2025).

Focus of the present study

Among the growing array of LLM-assisted topic modeling frameworks, TopicGPT (Pham et al., 2024) and LLoOM (Lam et al., 2024) stand out for their distinctive approaches to enhancing interpretability and conceptual rigor. While both leverage LLMs to support the generation and structuring of topics, they do so through fundamentally different philosophies and workflows.

TopicGPT: Prompt-Driven Topic Generation and Assignment

TopicGPT is built around a prompt-based, iterative refinement process that uses LLMs to generate, refine, and assign topics across a corpus (Pham et al., 2024). Unlike classical models that output topics as bag-of-words lists, TopicGPT produces natural language topic labels accompanied by concise

descriptions, yielding interpretable themes aligned with human categorizations. TopicGPT treats topic modeling as a prompt-driven, interpretable labeling process rather than a purely statistical decomposition, as it forgoes the “bag-of-words” paradigm in favor of human-readable topics defined by concise language, which prioritizes interpretable topic labels and descriptions. Its pipeline consists of three core stages: generation, refinement, and assignment.¹ This approach emphasizes human-readable output and encourages topic interpretability by treating LLMs as fluent generators of labeled topic descriptions. TopicGPT’s iterative prompt design also makes it easy to customize or adapt for various domains, which is valuable for applied communication research.

LLooM: Mixed-Initiative Concept Induction and Abstraction

LLooM is a concept induction framework that uses LLMs in a mixed-initiative loop to extract high-level concepts from unstructured text (Lam et al., 2024). It addresses a key shortcoming of conventional topic models: their tendency to focus on low-level keyword co-occurrences that are often too granular or incoherent for direct use in social science analysis. LLooM fundamentally reconceptualizes topic modeling by shifting from latent topic discovery to explicit concept building. The theoretical motivation is to bridge the gap between automated text analysis and the kind of high-level theorizing that human analysts perform. Traditional topic models require analysts to interpret cryptic word clusters post hoc, often struggling to map them onto theoretical constructs. LLooM inverts this relationship by producing interpretable, theory-friendly concepts, each representing a hypothesis about a theme in the data, described in expert-like language. Its pipeline consists of four stages: distill, cluster, synthesize, and score.² LLooM is explicitly designed to reduce the human burden of abstraction in qualitative analysis by transforming vague or overlapping themes into well-defined, theory-grounded concepts. The inclusion of document-level scores enables fine-grained, quantitative analyses of concept prevalence and distribution, which are typically missing in traditional or clustering-based models.

While both TopicGPT and LLooM use LLMs to enhance topic interpretability, their epistemological foundations diverge in meaningful ways. TopicGPT relies on LLMs as fluent generators of natural-language labels: it iteratively proposes, refines, and assigns topic descriptions directly from the text, treating the LLM as a high-capacity labeler and classifier. In this view, a

¹The workflow details are provided in Appendix E.

²The workflow details are also provided in Appendix E.

“topic” is essentially a human-readable summary that captures the dominant theme of a document or cluster of documents. In contrast, LLoOM treats LLMs not as label generators but as distillers and operationalizers of conceptual meaning. Rather than producing topic names directly, LLoOM extracts salient textual fragments, clusters them, and prompts the LLM to synthesize higher-level concepts with explicit definitions and inclusion criteria. These concepts are then applied back to the corpus through document-level scoring, and can be iteratively abstracted into broader conceptual categories. In short, TopicGPT generates topics as descriptive labels, whereas LLoOM induces structured concepts that more closely resemble theory-grounded codebook categories. These approaches also differ from earlier methods such as BERTopic or LDA. TopicGPT and LLoOM integrate LLMs directly into the topic creation process - either through prompt-driven topic generation (TopicGPT) or through concept synthesis and operationalization (LLoOM), yielding outputs that are both more interpretable and more aligned with researchers’ analytic expectations.

The present study focuses on TopicGPT and LLoOM because they represent two complementary paths forward in LLM-assisted topic modeling. By comparing these two approaches, this study aims to clarify how different LLM workflows support distinct analytical needs in topic modeling. Accordingly, we pose the following research questions:

- **RQ1:** How do LLM-assisted topic modeling approaches, i.e., TopicGPT and LLoOM, compare in terms of topic quality, coherence, diversity and interpretability?
- **RQ2:** What are the respective strengths and limitations of TopicGPT and LLoOM?
- **RQ3:** How can a practical framework be developed to guide researchers in selecting and applying LLM-assisted topic modeling methods effectively?

Method

Data

To ensure broad coverage across document length, topic diversity, and textual format, we utilized the following three datasets to address RQ1 and RQ2:

- The classic *20NewsGroup* dataset (*Post*), contains approximately 20,000 posts distributed across twenty topics. This dataset is commonly used

as a benchmark in natural language processing research, which features a diverse collection of informal textual content that reflects the style and format of online discussions, with varying document lengths (Mitchell, 1997).

- A collection of quasi-sentences from policy platforms of both Democratic and Republican parties (*Policy*). These statements were obtained from the Policy Agendas Project and represent condensed political positions articulated in party platforms from 1948 to 2020 (Wolbrecht et al., 2023a, 2023b). The dataset spans 21 policy topics and is primarily composed of sentence-length entries.
- The U.S. Congressional Bills dataset includes 47,973 bills introduced in the U.S. House of Representatives and Senate during the 115th to 117th Congresses (*Bill*). Curated by Hunter Heidenreich, this dataset contains formal legislative texts authored by the U.S. Congress. The documents are long-form and categorized into thirty-three distinct policy topics classified by <https://www.congress.gov/help/field-values/policy-area> (Heidenreich, 2023).

Given the computational cost and time associated with processing documents via LLMs, we employed a stratified sampling strategy to select 200 documents from each dataset, following prior work such as Lam et al. (2024).³ To ensure representativeness, we stratified samples primarily based on topic categories with all the topics involved in the original datasets. Additionally, for the *Policy* dataset, we included party affiliation (Democratic or Republican) as a stratification criterion. For the *Bills* dataset, stratification was further guided by congressional session (69 bills for 115th, 66 bills for 116th, and 65 for 117th). A comprehensive breakdown of the sampling distribution across these dimensions is provided in Appendix A. Table 1 presents a descriptive summary of the three sampled datasets.

Models

In this study, we employed four distinct topic discovery methods, LDA, BERTopic, TopicGPT, and LLoM, each paired with four LLMs: GPT-3.5-turbo, GPT-4o-mini, Gemini-1.5-pro, and Gemini-2.0-flash and one open weights model: Llama-3.1-8B-Instruct. The selected LLMs were chosen

³In the case of *LLoM*, increasing the number of input documents can quickly exceed the rate limits imposed by OpenAI’s API. As suggested by the original authors, although documents could be segmented into smaller batches, rate limitations continue to pose challenges during the scoring stage.

Table 1: Summary of sample datasets

Dataset	Documents	Format	Topics	Mean length
Post	200	A mixture of short- and medium-length user-generated content	20	151.73 words
Policy	200	Short quasi-sentences from party platforms	21	22.80 words
Bill	200	Long-form legislative documents	33	2268.74 words

based on a combination of performance, accessibility, and recency, as well as practical considerations related to API rate limits.⁴ As a classic and widely used baseline in topic modeling, we include LDA to anchor our comparison.⁵ We also adopted BERTopic as the other baseline method.⁶ While BERTopic traditionally returns a list of keywords per topic, we further adapted its output using prompt engineering based on the top 10 keywords and top 5 representative documents to ensure comparability across methods.⁷ For

⁴These models represent leading proprietary systems from two major developers, OpenAI and Google, and reflect a range of capabilities in terms of model size, inference speed, and cost. GPT-3.5-turbo and GPT-4o-mini offer strong performance with relatively low latency and cost, making them suitable for scalable experimentation. Gemini-1.5-pro and 2.0-flash, on the other hand, provide competitive alternatives with enhanced long-context handling and rapid response generation.

⁵We employed a comprehensive grid search approach to simultaneously optimize the number of latent topics (K) and the Dirichlet priors for document-topic density (α) and topic-word density (η), please see details in Appendix B.

⁶We implemented BERTopic using its standard pipeline with transformer-based embeddings, UMAP, and HDBSCAN. Document embeddings were generated with the default Sentence-Transformers model “all-MiniLM-L12-v2.” For vectorization and topic representation, we used a CountVectorizer with English stop-word removal, a minimum document frequency of two, and an n-gram range of one to three words, followed by refinement with the KeyBERTInspired model. We applied UMAP for dimensionality reduction with five nearest neighbors, cosine distance, and a fixed random seed, tuning the output dimensions to approximately 10–20 depending on the corpus. For clustering, we used HDBSCAN with a minimum cluster size of five and allowed the number of topics to be determined automatically. Parameters were selected through preliminary runs and qualitative inspection to preserve semantic coherence and topic diversity without overfragmentation or overcompression.

⁷Due to the length of documents in the bill dataset and the rate limits imposed by OpenAI’s models, we employed the “cl100k_base” tokenizer and limited the input to a maximum of 80,000 tokens (16385 tokens for GPT-3.5-turbo). For the other datasets, the full text of each document was processed without truncation.

TopicGPT and LLoM, all prompts used for each stage are documented in Appendix E.⁸

Evaluation metrics

Prior methodological reviews of topic modeling underscore that no single metric can capture all these dimensions reliably; instead, a combination of clustering/agreement metrics, coherence and diversity measures, embedding-based semantic metrics, and human validation is recommended (Meaney et al., 2023).⁹

Topic Alignment

When ground-truth labels are available, clustering alignment metrics are useful to quantify how well the predicted topics correspond to known categories. These metrics assess the agreement between predicted and true partitions, capturing both within-cluster homogeneity and cross-cluster separation. We use the following alignment metrics:

- **Harmonic mean purity:** We used the harmonic mean of topic purity and inverse purity to capture both the homogeneity of documents within each predicted topic and the consistency of ground-truth labels across predicted clusters. Higher harmonic mean purity indicates stronger alignment between predicted and actual topic structures.
- **Adjusted Rand Index (ARI):** We calculated ARI to measure the similarity between predicted and ground-truth cluster assignments based on pairwise label agreements, adjusted for chance. Higher ARI scores indicate better clustering performance, while values near zero suggest random assignment.
- **Normalized Mutual Information (NMI):** We employed NMI to quantify the mutual dependence between predicted and ground-truth topic assignments, normalized for cluster size. Higher NMI values indicate a greater degree of shared information between the two label sets.

⁸Although TopicGPT supports hierarchical topic modeling, including the identification of low-level subtopics under broader themes, we limited our analysis to high-level topics to maintain consistency across all three methods. Similarly, we didn't specify seed terms for LLoM to ensure the zero-shot learning nature in order to maintain the consistent comparison across methods.

⁹We presented a summary in the Appendix C that outlines each metric, its definition, scale, and interpretation guidelines to enhance clarity and readability

- **Semantic similarity (via SBERT embeddings):** We computed the cosine similarity between SBERT-encoded predicted topic labels and ground-truth labels using the pre-trained “all-MiniLM-L12-v2” model. Higher similarity values suggest greater semantic alignment between the predicted and actual topics.

Topic Coherence and Diversity

Yet clustering agreement does not guarantee semantic or interpretive quality: topics that align well with labels may still be semantically fuzzy or internally incoherent. For this reason, researchers routinely pair clustering metrics with intrinsic measures of coherence (to assess internal semantic consistency) and diversity (to assess topical coverage and distinctiveness). Embedding-based coherence or semantic-similarity metrics (e.g., using SBERT embeddings) further enhance this evaluation by moving beyond simple word-co-occurrence or frequency statistics to capture deeper semantic similarity (Yang & Kim, 2025). In particular, we use the following coherence and diversity metrics:

- **Intra-topic semantic coherence:** We assessed topic coherence by calculating the average pairwise cosine similarity of SBERT-based document embeddings within each predicted topic cluster. Higher coherence values indicate that documents grouped under the same topic are semantically similar, suggesting internally consistent clusters.
- **Top-1 coverage:** We measured the proportion of documents assigned to the most frequent predicted topic to assess topic dominance. Lower values indicate more balanced topic distributions, while higher values suggest over-reliance on a single topic.
- **Entropy of the topic distribution:** We computed the entropy of topic assignments to evaluate how evenly topics are distributed across the corpus. Higher entropy values reflect greater topical diversity, while lower values imply concentration in a few dominant topics.
- **Semantic diversity:** We calculated the mean pairwise cosine distance between all topic label embeddings to assess how semantically distinct the predicted topics are from one another. Higher values indicate broader topical coverage, while lower values suggest redundancy or overlap among topics.

Topic Vagueness

Besides, an important but often under appreciated dimension of topic quality is vagueness, i.e. whether topics or labels are overly generic or conceptually indistinct, failing to meaningfully differentiate themes in the corpus. In classical topic modeling, this problem arises when models produce topics dominated by high-frequency generic words (e.g., stopwords or common nouns), leading to topics that are trivially broad and less informative. Prior work has documented that standard coherence or likelihood metrics may not penalize such generic topics sufficiently, such that a model can attain high coherence while offering low utility for human interpretative purposes (Z. Li et al., 2024). We use the following three vagueness metrics:

- **Mean inverse document frequency (IDF):** We computed the average IDF of words in each predicted label based on the corpus vocabulary. Lower IDF values suggest the label consists of more frequent and thus potentially more generic terms.
- **Embedding centrality:** We calculated the cosine similarity between each label embedding and the mean embedding of all topic labels. Higher centrality values indicate that a label is semantically close to the average label, suggesting lexical or semantic generality.

Topic interpretability

Finally, human evaluation remains the gold standard for interpretability. Automated metrics can mislead: recent work shows that even strong coherence scores do not always correspond to human judgments about whether topics make sense, or whether topic assignments correctly reflect document content (Hoyle et al., 2021). Specifically, we trained four experienced researchers in communication and media studies to assess the quality of the generated topics. Specifically, the annotators evaluated two dimensions:

- **Label–topic matching:** Does the generated topic accurately reflect the ground truth label?
- **Text–topic matching:** Does the generated topic align with the actual document content?

For label–topic matching, scores were binary (0 or 1). For text–topic matching, scores were calculated as the ratio of correctly generated topics to the total number of generated topics. All ground-truth labels, original

documents, and generated topics were independently reviewed and annotated.¹⁰ During the training phase, each annotator evaluated 100 randomly selected documents.¹¹ To assess intercoder reliability, we used the generalized Jaccard similarity index to capture semantic overlap, rather than relying solely on exact agreement. This measure was applied to both single-label and multi-label cases. On average, the Jaccard similarity score was 0.74 for label–topic matching and 0.70 for text–topic matching across all annotators. Pairwise Jaccard similarity scores are reported in Appendix D. In the formal analysis, final scores reflect the mean values of both label–topic and text–topic matching for each subgroup.

Because each metric captures a different facet of topic quality, to evaluate topic model performance, we adopted a multi-dimensional framework encompassing topic alignment, coherence, diversity, and vagueness, incorporating both automated metrics and human annotation, as informed by prior literature (Lam et al., 2024; Pham et al., 2024; Rüdiger et al., 2022; Stambach et al., 2023). For the automated evaluation, all metrics were computed based on comparisons between the predicted topics and the ground-truth labels based on each subgroup.¹²

Results

Table 2 reports the distribution of label outputs. BERTopic consistently returned single labels but failed to label 72 documents across all LLM pairings. TopicGPT showed high variability; while GPT-4o-mini achieved consistent single-label performance (567 documents), Llama-3.1-8B-Instruct heavily favored multi-label outputs (469 documents). LLoM frequently produced multiple labels or zero-label outputs, particularly with Llama-3.1-8B-Instruct (344 zero-label documents).

As shown in Figure 1, the number of generated topics varied significantly by dataset length. BERTopic consistently under-segmented the semantic

¹⁰Documents without any generated topics were excluded from the analysis.

¹¹We have used specific examples to illustrate some of the rules of the annotation process in Appendix D.

¹²All the breakdown results could be found in Appendix C. While our evaluation strategy combines multiple complementary metrics, it remains an imperfect approximation of topic quality. Automated metrics, whether clustering-based (e.g., purity, ARI, NMI) or embedding-based (semantic similarity, coherence, diversity, vagueness), capture only a subset of what makes topics useful in real-world content analysis. As prior work argues, automated coherence and statistical measures often fail to align with human interpretability, especially for neural or embedding-based topic models (Z. Li et al., 2024; Tan & D’Souza, 2025). In other words, a model may achieve high coherence or clustering agreement but still produce topics that are too generic, overlapping, or semantically bland in ways that hinder substantive interpretation.

Table 2: Label distribution across methods and models

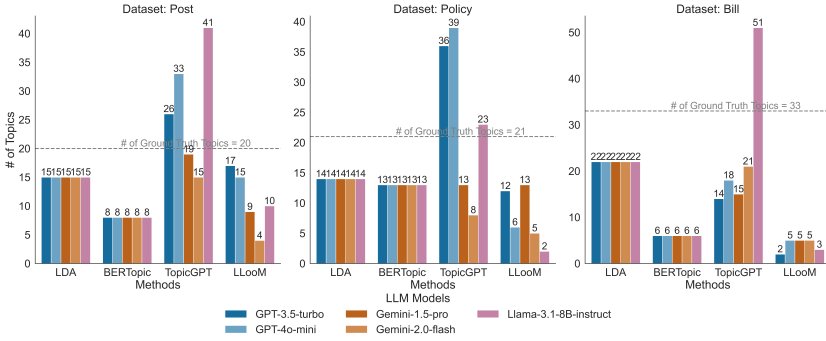
Models	Methods	Single label	Multiple labels	Zero label
GPT-3.5-turbo	LDA	600	0	0
GPT-4o-mini	LDA	600	0	0
Gemini-1.5-pro	LDA	600	0	0
Gemini-2.0-flash	LDA	600	0	0
Llama-3.1-8B-instruct	LDA	600	0	0
GPT-3.5-turbo	BERTopic	528	0	72
GPT-4o-mini	BERTopic	528	0	72
Gemini-1.5-pro	BERTopic	528	0	72
Gemini-2.0-flash	BERTopic	528	0	72
Llama-3.1-8B-instruct	BERTopic	528	0	72
GPT-3.5-turbo	LLoM	85	324	191
GPT-4o-mini	LLoM	179	343	78
Gemini-1.5-pro	LLoM	171	246	183
Gemini-2.0-flash	LLoM	224	77	299
Llama-3.1-8B-instruct	LLoM	132	124	344
GPT-3.5-turbo	TopicGPT	329	55	216
GPT-4o-mini	TopicGPT	567	32	1
Gemini-1.5-pro	TopicGPT	419	180	1
Gemini-2.0-flash	TopicGPT	489	96	15
Llama-3.1-8B-instruct	TopicGPT	125	469	6

space, collapsing complex corpora into fewer topics than the ground truth (e.g., only 6 topics for the Bill dataset versus 33 ground truth categories). Conversely, TopicGPT demonstrated a tendency toward granular segmentation, closely approximating or exceeding the ground truth count. For instance, on the Post dataset (20 ground truth topics), TopicGPT generated between 15 and 41 topics depending on the backend model. Notably, the open-weight Llama-3.1-8B-Instruct drove this granularity highest, producing 41 topics for Post and 51 for Bill within the TopicGPT framework, suggesting a propensity for fine-grained thematic distinction.¹³ LLoM functioned differently,

¹³In many applied communication research settings, overestimation of the number of topics is not inherently problematic and may even be desirable, as fine-grained topics can be meaningfully aggregated during substantive interpretation, whereas underestimation risks collapsing distinct discursive patterns that cannot be reliably recovered post hoc. We do not suggest that

favoring high-level abstraction; it produced the fewest concepts (2–5) for the Bill dataset, summarizing broad legislative themes rather than cataloging specific policy areas. For the evaluation metrics reported below, we present results in two parts: (1) single-label classification and (2) multi-label classification.

Figure 1: Number of topics generated by datasets, methods, and models.



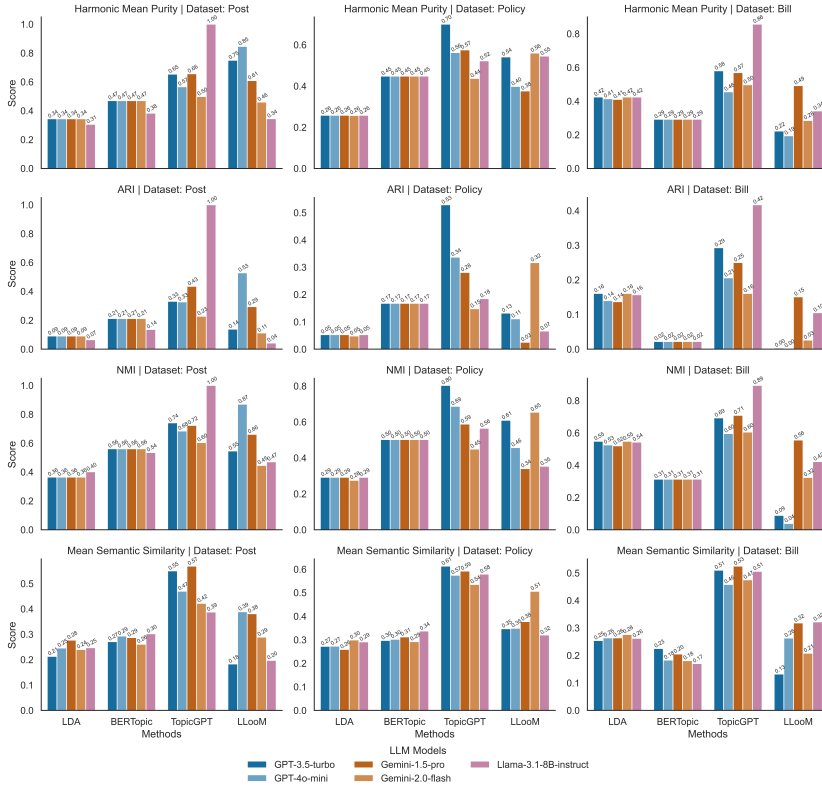
Single-label classification

Topic alignment

Figure 2 summarizes performance across harmonic mean purity, ARI, and NMI. Overall, alignment results generally favored TopicGPT, especially on the Policy and Bill datasets, though the pattern was not uniform across datasets or models. TopicGPT was the most consistent alignment-oriented framework overall, yet LLoOM with GPT-4o-mini achieved the highest purity on the Post dataset, and Llama-3.1-8B-Instruct produced the strongest Bill-dataset purity and NMI in specific TopicGPT configurations. Within TopicGPT, Gemini-1.5-pro achieved the highest overall purity on the Policy dataset (0.803). On the Bill dataset, however, the open-weights model Llama-3.1-8B-Instruct achieved the highest purity (0.857), substantially outperforming proprietary models such as GPT-3.5-turbo (0.580). Although TopicGPT was generally robust, the highest purity on the Post dataset was achieved by LLoOM with GPT-4o-mini (0.846), followed by LLoOM with GPT-3.5-turbo (0.750).

LLM-assisted methods resolve the well-known indeterminacy of topic number; rather, different levels of granularity can reveal different structures in the data, and LLM-based workflows offer alternative ways of navigating this trade-off through interpretability and conceptual abstraction rather than by identifying a single “correct” K.

Figure 2: Clustering and semantic evaluation metrics across datasets by methods and models for single-label documents.

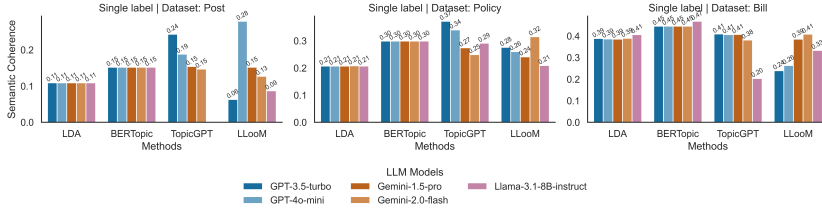


Model-specific differences also emerged across the alignment metrics. While the Gemini models slightly trailed the GPT models in purity, they performed strongly on NMI; notably, TopicGPT with Gemini-1.5-pro achieved a high NMI on the Bill dataset (0.709), though Llama-3.1-8B-Instruct recorded the highest overall NMI (0.894). Llama-3.1-8B-Instruct showed a distinct pattern: although it generally lagged behind proprietary models in semantic similarity, it excelled in specific configurations. For example, when paired with TopicGPT on the Policy dataset, it achieved a purity of 0.522 and semantic similarity of 0.579, performing competitively against Gemini-2.0-flash (purity 0.437, similarity 0.535).

Topic coherence and diversity

A clear trade-off emerged between coherence and diversity, as shown in Figures 3 and 4. Regarding coherence (Figure 3), TopicGPT with GPT-3.5-turbo proved highly effective, achieving the highest single-label coherence score for the Policy dataset (0.372), while GPT-4o-mini achieved 0.340 on the same corpus. For the Post dataset, LLoM with GPT-4o-mini led among generative methods with a coherence of 0.279. However, on the longer Bill dataset, the baseline BERTopic remained dominant due to its reliance on embedding-based clustering, though TopicGPT with Gemini-1.5-pro was competitive (0.407).

Figure 3: Intra-topics semantic coherence across datasets by methods and models for single-label documents.

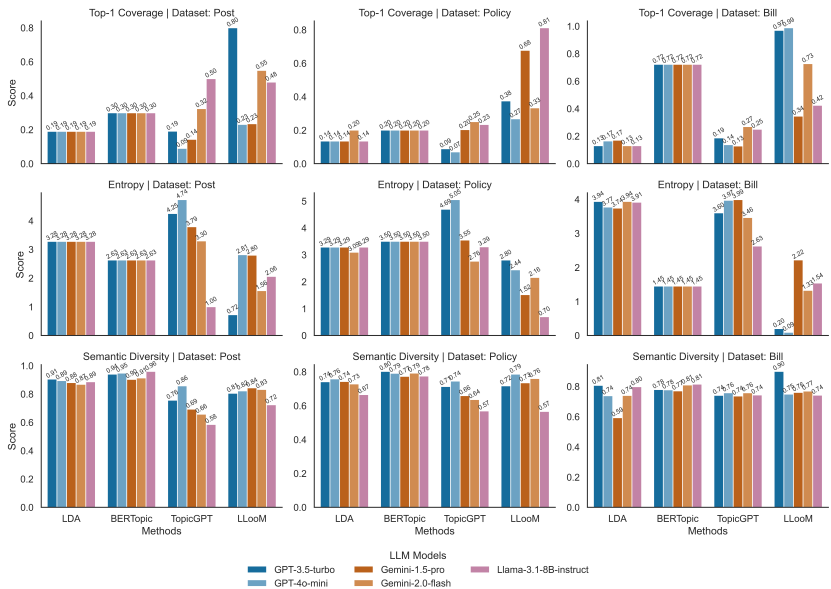


Meanwhile, regarding diversity (Figure 4), for the Post dataset, TopicGPT paired with GPT-4o-mini achieved the highest entropy score of 4.735, indicating a more balanced and diverse topic distribution compared to the baseline LDA (3.277). A similar pattern emerged in the Policy dataset, where GPT-4o-mini reached a peak entropy of 5.052. In the long-form Bill dataset, Gemini-1.5-pro slightly outperformed other generative models with an entropy of 3.992, while GPT-4o-mini followed closely at 3.970. Conversely, the Llama-3.1-8B-Instruct demonstrated significantly lower diversity in single-label settings; it recorded an entropy of only 1.000 on the Post dataset with a Top-1 coverage of 0.500, suggesting a heavy concentration on a narrow subset of themes.

Topic vagueness

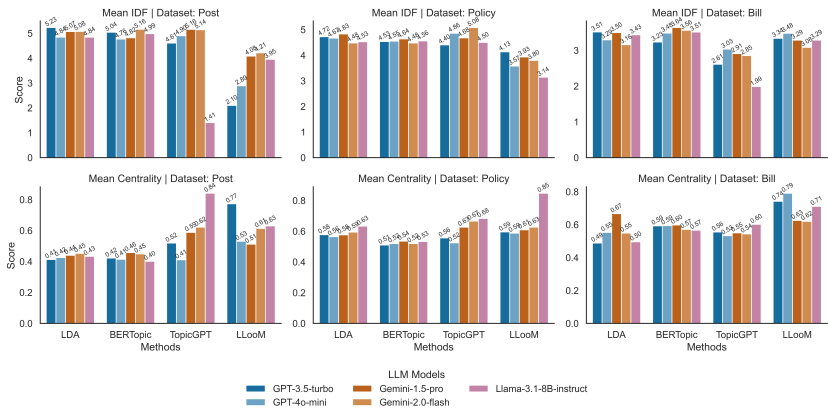
Figure 5 presents the trade-off between specificity and generality. TopicGPT with GPT-4o-mini consistently produced the most specific and least generic labels across all datasets. It recorded the lowest embedding centrality scores of 0.411 on the Post dataset and 0.524 on the Policy dataset, indicating that its labels were semantically distinct from the corpus average. In contrast,

Figure 4: Topic diversity metrics across datasets by methods and models for single-label documents.



Llama-3.1-8B-Instruct showed a stronger tendency toward semantic generality in single-label configurations. Its labels reached a centrality of 0.841 on the Post dataset, the highest (most vague) in that category, and 0.602 on the Bill dataset. Similarly, on the Policy dataset, Llama-3.1-8B-Instruct recorded a high centrality of 0.683, reflecting a broader, less specific naming convention compared to the GPT family.

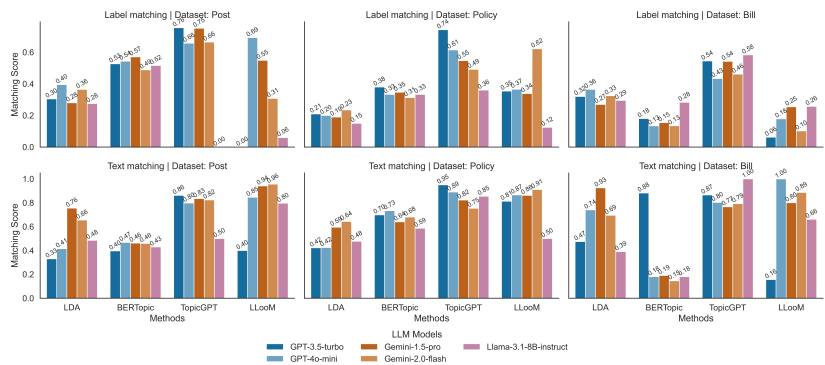
Figure 5: Topic vagueness metrics across datasets by methods and models for single-label documents.



Topic interpretability

For single-label documents, human annotation (Figure 6) revealed a consistent label-text gap where text-topic matching scores exceeded label-topic matching scores. TopicGPT consistently led in Label-Topic matching. Specifically, TopicGPT with GPT-3.5-turbo achieved the highest label matching score on the Post dataset (0.76). By contrast, LLoOM excelled in Text-Topic matching (relevance of the concept to the document content), with LLoOM with GPT-4o-mini achieving a perfect 1.00 text matching score on the Bill dataset. Llama-3.1-8B-Instruct struggled in this domain compared to proprietary models, particularly in label matching for the Policy dataset, though it maintained respectable text-matching performance on the Bill dataset when used with TopicGPT.

Figure 6: Matching scores by human annotation across datasets by methods and models for single-label documents.

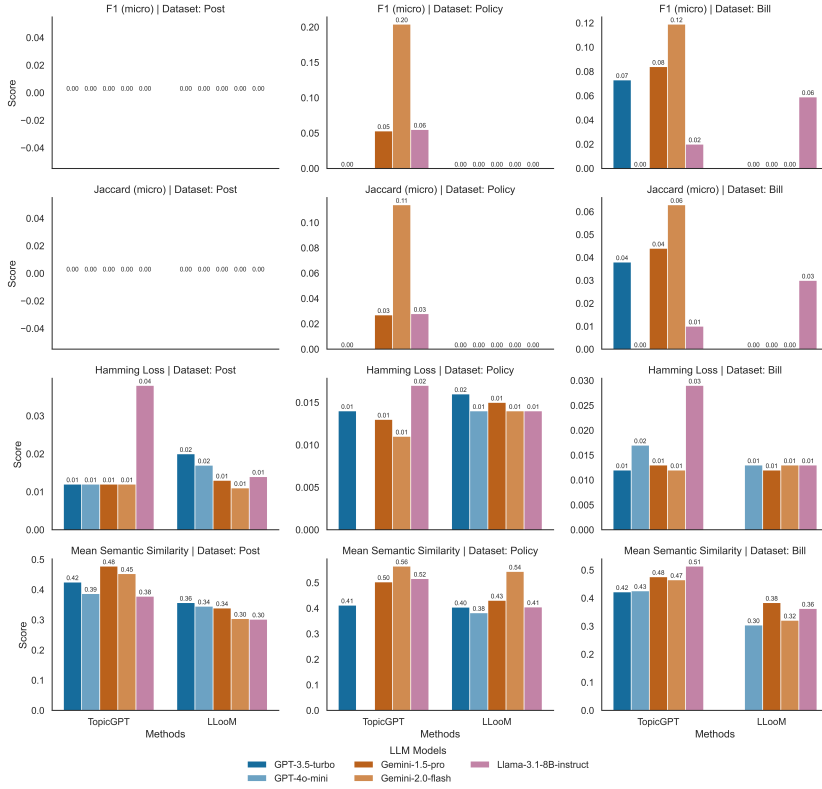


Multi-label classification

Topic alignment

Figure 7 presents multi-label evaluation metrics (excluding LDA and BERTopic). Multi-label alignment is a more challenging task, reflected in generally lower scores across all methods. Gemini-2.0-flash delivered the strongest Policy-dataset performance, but no single model family dominated uniformly across all datasets. Specifically, TopicGPT with Gemini-2.0-flash achieved the highest F1 score of 0.204 and a Jaccard index of 0.114 on the Policy dataset. Other models struggled to achieve meaningful alignment in multi-label settings. GPT-3.5-turbo and GPT-4o-mini both recorded F1 scores of 0.000 on the Post dataset. Similarly, GPT-3.5-turbo failed to align with ground-truth labels in the Policy dataset, yielding an F1 of 0.000. Llama-3.1-8B-Instruct provided more competitive multi-label results on the Bill dataset when paired with LLoOM, achieving an F1 score of 0.059, outperforming Gemini-1.5-pro and GPT-3.5-turbo in that specific configuration.

Figure 7: Clustering and semantic evaluation metrics across datasets by methods and models for multiple-label documents.



Topic coherence and diversity

In multi-label settings, TopicGPT and LLoOM showed comparable coherence (Figure 8). On the Post dataset, TopicGPT with GPT-3.5-turbo reached a coherence of 0.135, while TopicGPT with Gemini-2.0-flash led the Policy dataset with 0.285. For the Bill dataset, LLoOM with Llama-3.1-8B-Instruct achieved the highest multi-label coherence at 0.390.

Diversity metrics highlighted the strengths of concept induction and open-weight models in multi-label settings (Figure 9). LLoOM with GPT-3.5-turbo achieved an entropy of 4.170 on the Post dataset. Llama-3.1-8B-Instruct demonstrated exceptional coverage, producing the highest entropy distributions for the Policy (4.273) and Bill (5.618) datasets when paired with TopicGPT. These scores substantially exceeded those of proprietary models, such as Gemini-1.5-pro on the Policy dataset (3.229).

Figure 8: Intra-topics semantic coherence across datasets by methods and models multiple-label documents.

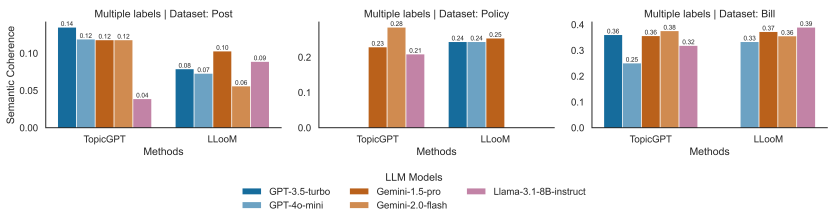
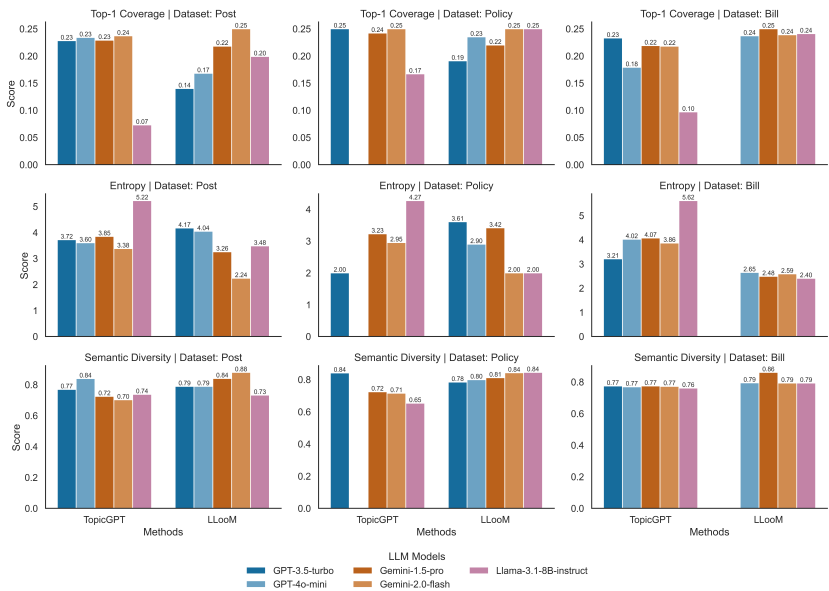


Figure 9: Topic diversity metrics across datasets by methods and models for multiple-label documents.

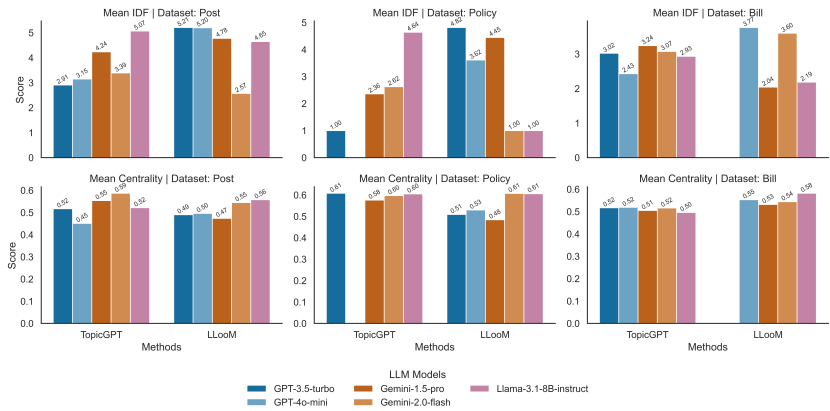


Topic vagueness

In the multi-label context (Figure 10), TopicGPT generally produced labels with higher centrality, indicating a tendency toward generic naming. However, Llama-3.1-8B-Instruct paired with TopicGPT on the Post dataset achieved a centrality of 0.522, which was competitive with GPT-3.5-turbo (0.517) and lower (more specific) than Gemini-2.0-flash (0.554). For the Policy dataset, centrality scores converged across models, with Gemini-2.0-flash (0.607) and Llama-3.1-8B-Instruct (0.605) producing similarly generic

labels compared to the more specific outputs seen in single-label tasks.

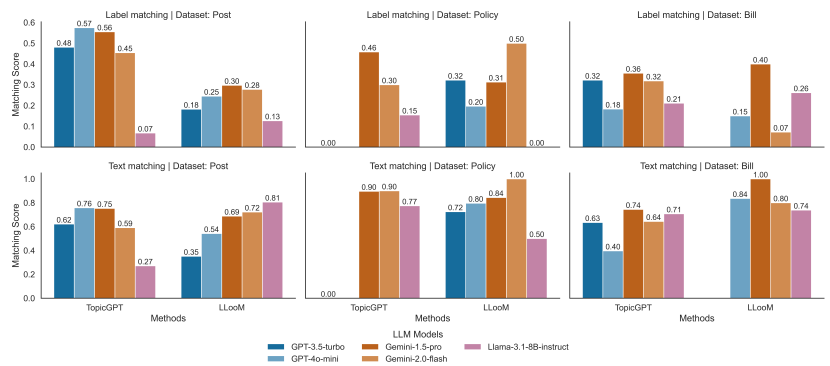
Figure 10: Topic vagueness metrics across datasets by methods and models for multiple-label documents.



Topic interpretability

Human evaluation for multi-label tasks (Figure 11) showed that TopicGPT with GPT-4o-mini achieved the strongest balance on the Post dataset with a Label-matching score of 0.57. LLoOM continued to dominate text relevance in complex tasks; LLoOM with Gemini-2.0-flash achieved a perfect 1.00 text-matching score for the multi-label Policy dataset.

Figure 11: Matching scores by human annotation across datasets by methods and models for multiple-label documents.



Qualitative evaluation

A close reading of the topic labels generated across the three datasets reveals a fundamental divergence in how TopicGPT and LLoM process textual meaning (see topic details in Appendix A). In particular, we focused on semantic abstraction: whether a method tends to generate (a) topical descriptors closely tied to literal domains (“topic categorization”), or (b) higher-level constructs that describe discursive functions, values, or latent frames (“concept induction”).¹⁴ This qualitative evaluation does not replace the technical evaluation; rather, it clarifies what some “misalignment” cases may represent when a method optimizes for a different level of meaning. While technical metrics penalize deviation from ground truth, a qualitative inspection suggests these deviations often represent valid, higher-level conceptual abstractions rather than classification errors.

In Post dataset, the models diverge in how they categorize social exchange. TopicGPT functions primarily as a categorizer of manifest subjects, producing domain-oriented labels (e.g., long-form topical inventories such as “Religion,” “Politics,” “Science”). These labels effectively index the subject matter of the discussion. Meanwhile, LLoM identifies latent discursive actions. Instead of simply labeling a discussion as a domain category, LLoM generates construct-like labels such as “Questioning Beliefs,” “Criticism of Authority,” and “Ethical Debates”.

For Policy dataset, the political-platform corpus illustrates a similar trade-off between specificity and framing. TopicGPT excels at replicating granular policy domains found in traditional codebooks, isolating specific issues such as “Employment Verification,” “Arms Control,” and other agenda-like categories. This aligns closely with the ground truth of political agendas and explains TopicGPT’s advantage on alignment-based metrics. LLoM, however, is designed to support concept induction: it aims to synthesize specific policy items into higher-level, value-laden constructs (i.e., frames) rather than only issue buckets. In other words, TopicGPT acts as a “splitter” that isolates explicit policy items, whereas LLoM is more of a “lumper” that (when successful) integrates multiple issue domains into broader interpre-

¹⁴We evaluate generated labels using a three-part rubric: (1) *Level of meaning* (manifest vs. latent): does the label primarily name an observable subject domain (e.g., “Employment,” “Arms Control”) or an abstract construct (e.g., critique, integration, empowerment)? (2) *Theoretical portability*: is the label readily reusable as a communication construct (e.g., skepticism, accountability, moral evaluation) rather than merely a topical tag? (3) *Operational transparency*: does the workflow provide explicit inclusion criteria (e.g., question-form rules used for document–concept matching) that make category boundaries auditable and support researcher validation?

tive constructs.

For Bill dataset, the congressional-bill corpus highlights the distinction between legislative vehicles and societal intent. TopicGPT consistently identifies manifest legislative topics or committee jurisdictions, outputting labels such as “Transportation and Public Works,” “Energy and Natural Resources,” and “Financial Services”. These labels accurately describe the type of bill and are well suited to agenda, jurisdiction, or domain attention analyses. LLoM, conversely, often attempts to capture societal outcomes or intent. This is consistent with LLoM’s mechanism: rather than treating labels as static buckets, it operationalizes concepts with explicit criteria and then evaluates document–concept matches in a dedicated scoring step.

Discussion

This study presents a systematic comparison of LLM-assisted topic modeling methods across three text datasets, especially moving from statistical inference (LDA/BERTopic) to semantic reasoning (TopicGPT/LLoM). Using a multi-dimensional evaluation framework with alignment, coherence, diversity, vagueness, and interpretability, we assessed the relative strengths and limitations of each method when paired with leading proprietary LLMs. In short, our results reveal that no single method dominates across all metrics. Rather, the choice of method imposes distinct trade-offs between semantic granularity, conceptual abstraction, and computational efficiency.

Quantitative synthesis: What the technical metrics collectively imply

Consistent with prior critiques of bag-of-words models (Chen et al., 2023; Egger & Yu, 2022), LDA provided a baseline of interpretability but struggled with semantic coherence especially in short or sparse texts where word co-occurrence signals are weak. In our evaluation, LDA’s topic alignment with known categories was generally low, reflecting the known limitation that semantically related terms can be split across different topics. LDA achieved moderate intra-topic coherence on longer documents (i.e., legislative bills), where abundant word co-occurrences help form coherent clusters. However, its topic diversity was limited as one or two broad topics often dominated the corpus, indicating lower entropy and coverage compared to other methods. The resulting topics were also vague in meaning, often composed of frequent general terms, and required subjective human labeling for interpretation.

BERTopic, while computationally efficient and offering high coherence

on long-form texts such as the Bill dataset, consistently under-segmented the topic space. Across all LLM backends, BERTopic showed lower topic alignment with ground-truth labels and reduced semantic diversity. Its reliance on static document embeddings and unsupervised clustering (UMAP + HDBSCAN) yielded stable clusters but often overly general topic representations, especially ill-suited for corpora with complex or nuanced themes. Coherence of BERTopic's topics was relatively high for long texts (since many documents were lumped into broad themes), but this came at the expense of interpretability and diversity. Even with prompt-based labeling assistance, the method's output quality remained heavily dependent on which representative documents were chosen for the LLM to label, leading to inconsistent naming. The vagueness of BERTopic's outputs was evident in its labels, which tended to use common terms. Overall, BERTopic's strength lies in speed and reproducibility, but its weak performance on alignment, diversity, and interpretability suggests caution in domains requiring conceptual precision. These outcomes echo recent assessments that purely unsupervised clustering approaches (even with embedding enhancements) often require substantial post-hoc interpretation and still underperform on fine-grained topic discovery (Chen et al., 2023; Lam et al., 2024).

Arguably, TopicGPT emerged as the most robust general-purpose framework in our study, providing a highly effective balance between semantic granularity and interpretive clarity. It consistently generated topic labels that closely aligned with ground-truth categories, particularly in structured political contexts like party platforms and legislative bills. It also yielded high intra-topic coherence; the documents grouped within its derived themes exhibited strong internal semantic similarity, suggesting the formation of cohesive and conceptually sound clusters. This strength was clearest when the method was paired with high-performance proprietary models, which scored well on both automated metrics and human evaluation. However, this strength involves a trade-off: some labels, while accurate, were somewhat broad. Nevertheless, TopicGPT often produced a more granular and differentiated topic structure than the baseline models, and in several dataset-model combinations it also distributed documents more evenly across topics, though this pattern was not uniform across all settings. From a practical standpoint, the prompt-generated labels and descriptive sentences significantly enhanced interpretability for human researchers. It is worth noting, however, that the flexibility of TopicGPT's prompt-driven approach results in less structured outputs compared to other methods, necessitating more intensive post-hoc data cleaning.

A sharp performance divergence emerged between single-label and multi-label contexts. While proprietary models achieved robust alignment in single-label tasks, performance degraded universally in multi-label settings, highlighting the difficulty of disentangling overlapping semantic themes in zero-shot environments. TopicGPT maintained a performance edge here, particularly when paired with the Gemini family, suggesting that its refinement stage helps preserve distinct topic boundaries when documents contain mixed signals. In contrast, open-weights models and concept-induction approaches struggled to assign multiple distinct labels accurately, often defaulting to a single dominant theme.

In contrast, LLooM offered a distinct approach centered on high-level concept induction rather than subject categorization. By design, LLooM prioritized the generation of fewer, more semantically distinct topics, making it an ideal tool for researchers seeking a broad overview of emergent thematic groupings. This was particularly evident in the analysis of long-form legislative texts, where LLooM produced highly interpretable, statement-like labels that captured the underlying intent or societal outcome of a document. Its focus on depth over breadth often resulted in strong text-topic matching. These conceptual strengths, however, are accompanied by significant operational and alignment challenges. Because LLooM often induces idiosyncratic labels, its direct alignment with manifest, pre-defined ground-truth categories was less consistent than that of TopicGPT. Furthermore, LLooM’s multi-stage iterative pipeline incurred substantially higher computational costs and was more susceptible to API rate limits, which may constrain its scalability for very large corpora. Ultimately, LLooM is best suited for exploratory tasks where identifying novel theoretical constructs is prioritized over the precise reproduction of literal subject labels.

Qualitative synthesis: From topic categorization to concept induction

A critical finding from our human annotation is the consistent discrepancy between label accuracy and text relevance. Across all models, annotators rated the relevance of the topic to the text significantly higher than the accuracy of the label itself. This suggests that while LLMs are adept at identifying the underlying latent themes of a document (text matching), they struggle to compress that semantic richness into a concise 2-3 word label at the translation layer (label matching). This compression loss might highlight a limitation in current zero-shot prompting strategies. This misalignment underscores the importance of human interpretative oversight. Automated

metrics alone cannot capture the full validity of topics, as what looks aligned quantitatively might still require qualitative verification to ensure the topic truly represents a coherent concept in context. Our findings reinforce that LLM-assisted tools should be viewed as augmented aids for expert analysts, not replacements for them. In practice, a hybrid approach is advisable: researchers can use these models to generate candidate topics and labels, but should iteratively refine and validate them against domain knowledge and through reading sample texts (Dai et al., 2023). Encouragingly, both TopicGPT and LLoM have configuration options (e.g. seeding with key terms or providing few-shot examples) that could further improve alignment and interpretability, bridging some gaps when expert input is incorporated. Future work can explore such semi-supervised refinements, but the current results already demonstrate that expert judgment remains essential for interpreting and validating topics, echoing long-standing recommendations in computational social science (Chang et al., 2009; Nelson, 2020).

The qualitative evidence challenges reliance on purely technical evaluation metrics. While our quantitative analysis showed TopicGPT achieving significantly higher alignment in structured policy contexts (e.g., a peak purity of 0.803 on the Policy dataset), the qualitative analysis suggests that “misalignment” in LLoM can reflect a shift in ontological focus: from Topic Categorization (manifest subjects) to Concept Induction (latent constructs). This distinction is methodologically grounded in how the models operate. TopicGPT functions as a deductive-style coder: it generates a list of labels and assigns documents into a static set of topic buckets, mimicking a human coder applying a pre-defined codebook of manifest subjects. This is supported by its low embedding centrality (0.411 on the Post dataset), which indicates a high degree of naming specificity for manifest subjects. LLoM, in contrast, is designed to function more like an inductive concept builder. Crucially, this is not only a labeling difference but a structural one. As implemented in our workflow, LLoM does not simply assign a label; it generates explicit inclusion criteria for each concept and then performs a scoring step. In this step, the model explicitly asks whether a document matches a concept definition, treating the concept as a hypothesis about meaning to be tested against the text. This concept matching process helps produce groups that are theoretically coherent constructs defined by shared meaning, rather than only clusters of co-occurring keywords.

This inclusion-criteria logic can be expressed as a concept prompt in the model output of LLoM, which means a concrete question that makes the concept's boundary conditions transparent. For example, a concept about

public-sector involvement could be operationalized by a prompt such as: Does the text discuss federal government support or involvement in various sectors such as arts, education, or infrastructure?” This kind of criteria prompt aligns with the LLoOM Score” template used in our implementation, which converts a concept into a prompt pattern and asks the model whether the document is primarily about that concept, with a rationale. In communication research terms, this criteria-based scoring provides interpretive scaffolding that can support construct validity: researchers can inspect, refine, and contest the criteria rather than relying only on opaque label names.

Hypothetically, one might argue that TopicGPT’s assignment stage performs a similar function to LLoOM’s score stage, as both map documents to themes. However, a closer inspection reveals a methodological divergence. TopicGPT’s assignment process asks the model to classify a document into a static list of generated labels, treating topics as fixed buckets, which is consistent with deductive coding where the goal is clean categorization. In contrast, LLoOM’s score function emphasizes criteria-based evaluation: it operationalizes a concept via explicit prompts and evaluates documents against those conceptual rules. The result is not only a label but an auditable decision logic for why a document fits a construct.

Cost, openness, and reproducibility considerations

The practical viability of LLM-assisted topic modeling must be weighed against computational cost and openness. In our experiments, the proprietary LLM APIs incurred non-trivial costs: depending on document length and the number of topics, a single full run ranged from roughly \$0.50 up to \$2.50 in API charges. For small-scale studies (like ours with a few hundred documents), these costs are manageable. However, scaling up to thousands of documents or using more complex multi-step prompts would multiply the expense significantly. Such costs can be prohibitive for researchers without dedicated funding and raise concerns about inequitable access to advanced methods. Beyond institutional disparities, these cost and infrastructure demands may also deepen global inequalities in computational communication research, as scholars working in lower-resourced or internationally uneven funding environments may face greater barriers to accessing proprietary LLM pipelines.

Meanwhile, reliance on closed-source APIs poses challenges for reproducibility and open science. Therefore, the inclusion of Llama-3.1-8B-Instruct provides a critical benchmark for current state-of-art open-weight models’

capabilities in zero-shot topic modeling, which reveals a distinct trade-off between semantic diversity and structural adherence. While the model lagged behind proprietary counterparts in ground-truth alignment and incurred significant latency during generative tasks, it unexpectedly outperformed them in thematic distinctiveness. Unlike RLHF-tuned proprietary models that tended to converge on broad, generic labels, Llama-3.1-8B-Instruct appeared less prone to mode collapse in some configurations, producing especially high-entropy topic distributions in multi-label settings, although its semantic vagueness varied across datasets. This suggests that while smaller open models may struggle with the strict instruction-following required for standardized categorization, their tendency to generate granular, idiosyncratic outputs makes them potentially superior for purely exploratory tasks where discovering niche themes is prioritized over standardization. However, our runtime analysis revealed significant latency issues for Llama-3.1 during the TopicGPT refinement and assignment phases. This suggests that smaller, open-weights models may struggle with the long-context dependencies and complex reasoning required for zero-shot topic assignment, resulting in generation bottlenecks.

Moreover, performance is heavily constrained by hardware infrastructure, particularly regarding memory limitations for handling long contexts — notably, the analysis in this study was conducted on NVIDIA A100 GPUs (40GB VRAM). While High-Performance Computing (HPC) resources are recommended to mitigate these constraints, they may not be universally accessible to researchers. As noted by Chae and Davidson (2025), although smaller models can be competitive, they often require instruction-tuning or few-shot examples to match the zero-shot reasoning capabilities of larger proprietary systems. Another potential approach will be leveraging domain-specific pre-trained models for specific subtasks, such as pre-trained entailment-based models for topic validation Burnham et al. (2024) and Kubli (2025). These options may offer competitive performance while improving accessibility and reproducibility for researchers operating with limited computational resources.

Conclusion and practical guidance

Our comparative analysis speaks to broader debates on advancing topic modeling with AI. The findings align with recent critiques in the literature Chen et al. (2023) about the interpretability limits of traditional topic models and provide concrete evidence that LLM-assisted methods can mitigate some of these issues. In line with Bail (2024) vision of generative AI augment-

ing social science, we show that LLM-based topic modeling can indeed yield more interpretable and semantically meaningful themes than classic LDA and BERTopic, thereby potentially improving the validity of computational text analysis in communication research. However, another implication concerns determinism and variability. Non-determinism is a concern in traditional topic modeling techniques, and the introduction of LLMs is not a remedy. LLM-assisted pipelines may even introduce additional sources of variability. Our results should consequently be interpreted as comparative benchmarks that highlight relative performance patterns, rather than as fully reproducible or deterministic outputs. In fact, LLM-assisted pipelines can introduce new sources of stochastic variation (e.g., randomness in prompt responses in TopicGPT or API instabilities in LLoM). As shown in Appendix F, TopicGPT demonstrated relatively higher stability in both topic granularity and semantic consistency, with low variance in the number of topics generated and a stronger average semantic overlap score (0.78). LLoM exhibited greater linguistic variability, yielding a more moderate semantic overlap (0.54). While TopicGPT in particular offers stronger reproducibility for applications requiring tighter categorization in our case, our output should be interpreted as comparative benchmarks highlighting relative performance patterns rather than exact reproducible partitions. This variability makes tool selection consequential: methods differ not only in average performance but also in the stability and epistemic level of meaning they tend to produce.

For communication scholars aiming to apply LLM-assisted topic modeling, our results offer concrete guidance on choosing the appropriate tool for the task at hand. TopicGPT emerges as a strong general-purpose choice when the research requires interpretable, fine-grained topics and the texts are short to mid-length. It produced more topics with greater topical granularity and semantic alignment than the keyword-based approaches, making it ideal for analyzing collections like social media posts, press releases, or speeches where nuance is important. If the goal is to map texts to predefined categories or to achieve high alignment with known issues, TopicGPT is recommended given its performance on label alignment and coherence. LLoM, by contrast, is advantageous when a broader overview of themes is sufficient or desired. In contexts where researchers seek a concise set of overarching themes, such as summarizing the main issues in a large corpus of policy documents or identifying the dominant frames in legislative debates, LLoM's ability to generate distinct and generalizable topics is valuable. Our study indicates that LLoM yields fewer topics that

nevertheless capture diverse concepts, which can simplify interpretation for high-level analysis. However, users of LLoOM should be mindful of its computational requirements and potential output variability. For projects with very large corpora or real-time analysis needs, the cost and rate-limit constraints of LLoOM (using current proprietary models) might outweigh its benefits (especially on the synthesize stage). In such cases, one might consider using TopicGPT with an open-weight model or limiting the scope of LLoOM's analysis (e.g., analyzing a representative sample rather than an entire big dataset). In other words, this distinction yields a clear decision pathway. When the research question asks "What topics are on the agenda?" (e.g., distinct policy domains), TopicGPT is well suited due to its precision in capturing manifest subjects. However, when the research question asks "What representational practices, frames, or discursive logics are being used?", LLoOM may provide an advantage because it is built to induce latent concepts and validate them via document–concept matching. Put differently, when the research goal is manifest agenda mapping or codebook-consistent categorization, TopicGPT is the more reliable choice; when the goal is latent construct discovery (e.g., frames or discursive logics), LLoOM's concept-induction design may be advantageous.

This framing also clarifies how to interpret the underperformance under alignment-based metrics. Importantly, we do not claim that concept induction is universally superior. By contrast, the lower performance of LLoOM on technical label alignment metrics should not automatically be interpreted as model failure; as our results show, LLoOM leads in text relevance (text-matching), suggesting it understands document content even when it deviates from ground-truth labels. As illustrated by the Post dataset examples in Appendix A, LLoOM's "errors" can be conceptually rich abstractions (e.g., "Questioning Beliefs") that offer theory-relevant perspectives not captured by manifest domain labels. Consequently, the choice between TopicGPT and LLoOM is not merely computational but epistemological: it hinges on whether the study requires reproducing literal topical categories or extracting latent concepts. We therefore urge scholars to evaluate tools not on a single leaderboard of alignment, but on their fitness for the theoretical level of meaning required by their research question. To translate this distinction into practical guidance, Table 3 summarizes a decision framework for selecting among LLM-assisted topic modeling approaches based on research scenario, data characteristics, and resource constraints.

Meanwhile, traditional methods still have their place in certain scenarios. While LDA and BERTopic are fast, easily deployable, their performance trade-

offs are significant. If a researcher's priority is computational efficiency or transparency, these methods can provide a starting point. But our empirical findings urge caution: both LDA and BERTopic struggled with interpretability (producing vague topics) and missed many nuanced distinctions that the LLM-assisted methods captured (which might be improved with larger number of representative documents). Thus, if used, traditional models should be followed by careful human interpretation and possibly augmented with LLM-based labeling to improve the output. In domains requiring high conceptual clarity and alignment with theory (e.g., mapping discourse to theoretical constructs or policy codes), the generative approaches are generally more reliable. In particular, BERTopic or LDA may be inappropriate when the text data spans numerous fine-grained categories, as they tend to merge topics or lose detail in such cases. Finally, regardless of method, we recommend scholars employ a robust evaluation routine: quantitative metrics (for alignment, coherence, etc.) should be complemented by human evaluation of topic meaningfulness, and any surprising model outputs should be cross-checked with the source documents.

In summary, no single topic modeling method is universally best – each involves trade-offs. Researchers should align their choice of tool with their data characteristics and research goals. To aid in this decision, we provide a decision framework distilled from our study's insights. Table 3 only serves as a practical reference for selecting and evaluating LLM-assisted topic modeling techniques in communication research, ensuring that users can leverage these emerging tools effectively while being aware of their limitations.

Table 3: Decision framework for selecting LLM-assisted topic modeling methods

Research Scenario	Recommended Method	Practice Guidance & Limitations
Confirmatory analysis (Reproducing known categories; coding tasks requiring high precision)	TopicGPT	Rationale: Achieves the highest alignment with human-defined ground truth; mimics iterative human coding logic effectively. Limitation: May over-segment topics and produce inconsistent label granularity or format (output structure) across models. Tip: Validate labels using the generated descriptions and representative quotes. Performs best on short-to-medium length texts.
Exploratory theory building (Inductive discovery of abstract or emergent concepts)	LLoM	Rationale: Excels at “concept induction,” producing semantically distinct and abstract themes rather than simple keyword clusters. Limitation: High computational cost (especially during scoring) and lower precision in alignment with manifest ground-truth labels, even when document-level text relevance is high. Tip: Use for identifying novel theoretical angles, but validate concept boundaries manually.
Large-scale content analysis (First-pass screening or clustering of very large corpora)	BERTopic	Rationale: Highly scalable and computationally efficient; provides stable, coherent clusters for long-form text. Limitation: Significant tendency to under-segment complex topics; lacks the semantic reasoning capabilities of generative models. Tip: Use as a filtering step. Refine coarse clusters via sub-topic modeling or human review.
Resource-constrained (No API budget)	TopicGPT (Open-weights)	Rationale: Eliminates reliance on proprietary APIs; ensures cost-free reproducibility. Limitation: Significantly higher latency (generation bottlenecks) and lower zero-shot performance compared to proprietary models. Tip: Requires robust hardware (HPC/GPU). Consider few-shot prompting to improve instruction-following.

Limitation

Despite the methodological breadth of this comparative evaluation, several limitations should be acknowledged. First, both traditional and LLM-based topic modeling approaches face meaningful reproducibility challenges. Although we fix random seeds where possible, classical models such as LDA

and BERTopic can still yield different topic solutions across runs, particularly with sparse or noisy corpora. LLM-assisted methods introduce additional layers of non-determinism: API-level randomness, stochastic decoding, and the ongoing server-side updates typical of large proprietary models. These factors constrain exact replication as what we can see in a small-scale stability test in Appendix F.

Second, prompt design remains an important but unresolved source of variability. We intentionally used the original prompts provided by TopicGPT and LLoM to maintain fair comparability with their published configurations and to avoid over-tuning. However, we did not conduct a systematic prompt-optimization procedure or test alternative formulations. Emerging research demonstrates that even subtle prompt modifications can affect LLM outputs. We highlight this as a direction for future work and recommend that researchers conduct prompt-sensitivity analyses when resources permit.

Third, our evaluation relied on targeted subsamples (200 documents per corpus) to keep the full 5-model * 4-method * multi-metric design computationally tractable. Although these datasets reflect common political communication genres, they are smaller than the large-scale corpora typically analyzed in computational communication research. Because our comparative design relied on relatively small subsamples, the performance of classical and clustering-based methods such as LDA and BERTopic may be somewhat conservative relative to their behavior on much larger corpora, where richer co-occurrence and embedding structure can stabilize topic discovery. At the same time, this constraint also highlights a practical advantage of LLM-assisted approaches, which may remain useful in research settings where labeled data are unavailable and corpus size is comparatively modest. We acknowledge that LLM-assisted pipelines, especially those involving proprietary APIs, may be prohibitively expensive for very large corpora or for research groups with limited computational resources. We have introduced open-weights Llama-3.1-8B-Instruct and discuss how researchers can combine open models with batched scoring, sub-sampling strategies, or BERTopic + LLM labeling to improve feasibility at scale. Similarly, while our addition of LDA strengthens benchmarking, a full grid search across alternative classical and neural topic models was beyond the scope of the present study.

Fourth, although we expanded and clarified our metric explanations, some automated metrics, such as semantic diversity or embedding-based coherence, remain imperfect proxies for human interpretability. We there-

fore incorporated human-centered evaluation with four trained annotators. Traditional word-intrusion tasks, however, were not feasible for LLoM and TopicGPT, which do not return probability-ranked top-n word lists. We acknowledge that adapting intrusion-style evaluation to generative topic descriptions would be a valuable direction for future methodological research.

Fifth, our study focuses on zero-shot topic induction and categorization, rather than guided topic modeling or few-shot learning. While this choice reflects common real-world use cases, it limits our ability to evaluate whether LLoM or TopicGPT can detect complex communication constructs without additional supervision. Future work could incorporate domain-specific seeding, or hybrid human-in-the-loop annotation with a few examples to evaluate these capabilities more fully.

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